1. The angles of a triangle are in A.P. If b : c = $\sqrt{3}/\sqrt{2}$ then angle A is equal to: **NIMCET 2007**
(a)75° (b)105° (c)30° (d)45°

2. mth term of H.P. is n and its nth term is m, then find $(m+n)^{th}$ term? **NIMCET 2007**
(a)$\frac{mn}{m+n}$ (b)$\frac{m+n}{mn}$ (c)$\frac{m-n}{mn}$ (d)none

3. The ages of students in a group are in A.P. If the youngest is 5 years old, and the oldest is 15 years old, then the average age of the group is : **NIMCET 2007**
(a)5 (b)15 (c)10 (d)can't say

4. Suppose a, b, c are in AP with common difference d. Then, $e^{1/c}, e^{b/ac}, e^{1/a}$ are in **NIMCET 2008**
(a)AP (b)GP (c)HP (d)None

5. If $H_1, H_2, \ldots, H_n$ are n harmonic means between a and b, $a \neq b$, then the vector of $\frac{H_1+a}{H_1-a}+\frac{H_n+b}{H_n-b}$ is equal to **NIMCET 2008**
(a)$n+1$ (b)$n-1$ (c)2n (d)$2n+3$

6. The mean of first n natural numbers is equal to $\frac{n+7}{3}$, then 'n' is equal to **NIMCET 2008**
(a)9 (b)10 (c)11 (d)12

7. If a, b, c are in Ap, p, q.r in HP and ap, bq, cr are in GP, then, $\frac{p}{r}+\frac{r}{p}$ is equal to **NIMCET 2010**
(a)$\frac{a}{c}-\frac{c}{a}$ (b)$\frac{a}{c}+\frac{c}{a}$ (c)$\frac{b}{q}-\frac{a}{p}$ (d)$\frac{b}{q}+\frac{a}{p}$

8. The sum of $11^2+12^2+\cdots+30^2$ **NIMCET 2011**
(a)8070 (b)9070 (c)1080 (d)9700

9. If three positive real number a, b, c $(c > a)$ are in HP, then $\log(a+c)+\log(a-2b+c)$ is **NIMCET 2011**
(a)$2\log(c-b)$ (b)$2\log(a+c)$
(c)$2\log(c-a)$ (d)$\log a+\log b+\log c$

10. If the mean of the squares of first n natural numbers be 11, then n is equal to **NIMCET 2011**
(a)-13/2 (b)11 (c)5 (d)4

11. If H the harmonic mean between P and Q, then $\frac{H}{P}+\frac{H}{Q}$ is **NIMCET 2012**
(a)2 (b)$\frac{P+Q}{Q}$ (c)$\frac{PQ}{P+Q}$ (d)None

12. If the real number x when added to its inverse gives the minimum value of the sum, then the value of x is equal to **NIMCET 2012**
(a)-2 (b)2 (c)1 (d)-1

13. In a GP consisting of positive terms, each term is equals the sum of the next two terms. Then, the common ratio of the GP is **NIMCET 2013**
(a)$\frac{(1-\sqrt{5})}{2}$ (b)$\frac{(\sqrt{5})}{2}$ (c)$\sqrt{5}$ (d)$\frac{(\sqrt{5}-1)}{2}$

14. The sum of n terms of an arithmetic series is 216. The value of the first term is n and the value of the nth term is 2n. the common difference d is **NIMCET 2013**
(a)1 (b)$\frac{2}{3}$ (c)$\frac{3}{2}$ (d)$\frac{12}{11}$

15. The sum of integers between 200 and 400, that are multiples of 7 is **NIMCET 2013**
(a)8729 (b)8700 (c)8972 (d)8279

16. The sum of 20 terms is series $-1^2+2^3-3^2+4^2-\cdots$ is **NIMCET 2013**
(a)180 (b)200 (c)210 (d)220

17. Let $f(x)$ be a polynomial function of second degree and $f(1)=f(-1)$. If a, b and c are in AP, then $f'(a), f'(b)$ and $f'(c)$ are in **NIMCET 2013**
(a)GP (b)HP (c)AGP (d)AP

18. The value of $9^{\frac{1}{3}}\cdot 9^{\frac{1}{9}}\cdot 9^{\frac{1}{27}}$....... Is **NIMCET 2013**
(a)3 (b)6 (c)9 (d)None

19. If a, b and c are in geometric progression, then $\log_{ax} x, \log_{bx} x$ and $\log_{cx} x$ are in **NIMCET 2015**
(a)arithmetic progression
(b)geometric progression
(c)harmonic progression
(d)arithmetic-geometric progression

20. The value of the sum $\frac{1}{2\sqrt{1}+1\sqrt{2}}+\frac{1}{3\sqrt{2}+2\sqrt{3}}+\frac{1}{4\sqrt{3}+3\sqrt{4}}+\cdots+\frac{1}{25\sqrt{24}+24\sqrt{25}}$ is **NIMCET 2015**
(a)$\frac{9}{10}$ (b)$\frac{4}{5}$
(c)$\frac{14}{15}$ (d)$\frac{7}{15}$

21. If $a_1, a_2 \dots . a_n$ are in AP and $a_1 = 0$, then the value $\left(\frac{a_3}{a_2} + \frac{a_4}{a_3} + \cdots \frac{a_n}{a_{n-1}}\right) - a_2\left(\frac{1}{a_2} + \frac{1}{a_3} + \cdots \frac{1}{a_{n-2}}\right)$ is equal to **NIMCET 2016**

(a)$(n-2) + \frac{1}{(n-2)}$ (b)$\frac{a}{(n-2)}$
(c)$(n-2)$ (d)$n - \frac{1}{(n-2)}$

22. The sum of expansion $\frac{1}{\sqrt{1}+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \cdots + \frac{1}{\sqrt{80}+\sqrt{81}}$ is **NIMCET 2016**

(a)7 (b)8 (c)9 (d)10

23. The harmonic mean of two numbers is 4. Their arithmetic mean A and the geometric mean G satisfy the relation $2A + G^2 = 27$. Then, the two numbers are **NIMCET 2017**

(a)2 and 3 (b)6 and 3
(c)5 and 7 (d)4 and 1

24. Three positive numbers whose sum is 21 are in arithmetic progression. If 2, 2, 14 are added to them respectively, then resulting numbers are in geometric progression. Then, which of the following is not among the three numbers? **NIMCET 2017**

(a)25 (b)13 (c)1 (d)7

25. If $a_1, a_2, a_3 \dots a_n$ are positive real numbers whose product is fixed number C, then the minimum value of $a_1 + a_2 \dots + 2a_n$ is **NIMCET 2018**

(a)$n(2C)^{1/n}$ (b)$(n+1)C^{1/n}$
(c)$2nC^{1/n}$ (d)$(n+1)(2C)^{1/n}$

26. Sum of infinity of geometric progression is twice the sum of first two terms. Then, what are the possible values of the common ratio? **NIMCET 2018**

(a)$\pm\frac{1}{\sqrt{2}}$ (b)$\pm\frac{1}{2}$ (c)$\pm\frac{1}{\sqrt{3}}$ (d)$\pm\frac{1}{3}$

27. Suppose that m and n are fixed numbers such that the mth term a_m is equal to n and nth term a_n is equal to $m(m \neq n)$, then the $(m+n)^{th}$ term is **NIMCET 2018**

(a)$(m+n)/mn$ (b)$mn/(m+n)$
(c)$(m+n)/n$ (d)$(m+n)/m$

28. If the mean of the squares of first n natural numbers be 11, then is equal to **NIMCET 2018**

(a)$\frac{-13}{2}$ (b)11 (c)5 (d)4

29. If $a, a_1, a_2, a_3 \dots . . a_{2n-1}$, b are in AP, $a, b_1, b_2, b_3 \dots b_{2n-1}$, b are in GP and $a, c_1, c_2, c_3, \dots . c_{2n-1}$, b are in HP, where a, b are positive, then the equation $a_n x^2 - b_n x + c_n = 0$ has its roots **NIMCET 2019**

(a) real and equal
(b) real and unequal
(c) imaginary
(d) one real and one imaginary

30. If a, b, c are in GP and $\log a - \log 2b, \log 2b - \log 3c$ and $\log 3c - \log a$ are in AP, then a, b, c are the lengths of the sides of a triangle which is **NIMCET 2019**

(a) acute angled (b) obtuse angled
(c) right angled (d) equilateral

31. If $x, 2x+2, 3x+3$ are the first three terms of a geometric progression, then 4th term in the geometric progression is **NIMCET 2019**

(a)-13.5 (b)13.5 (c)-27 (d)27

32. The sum of infinite terms of decreasing GP is equal to the greatest value of the function $f(x) = x^3 + 3x - 9$ in the interval [-2, 3] and difference between the first two terms is $f'(0)$. Then, the common ratio of the GP is **NIMCET 2019**

(a)$-\frac{2}{3}$ (b)$\frac{4}{3}$ (c)$+\frac{2}{3}$ (d)$-\frac{4}{3}$

33. If $\sum_{i=1}^{n} x_i = 80$ and $\sum_{i=1}^{n} x_i^2 = 400$, then a possible value of n among the following is **NIMCET 2019**

(a)9 (b)12 (c)15 (d)18

34. An arithmetic progression has 3 as its first term. Also, the sum of the first 8 terms is twice the sum of the first 5 terms. Then, what is the common difference? **NIMCET 2020**

(a)3/4 (b)1/2 (c)1/4 (d)4/3

35. If $H_1, H_2 \dots . . H_n$ are n harmonic means between a and b $(b \neq a)$, then $\frac{H_1+a}{H_1-a} + \frac{H_n+b}{H_n-b} = ?$ **NIMCET 2021**

(a)2n (b)$n+1$ (c)$n-1$ (d)$2n+1$

36. The four geometric means between 2 and 64 are **NIMCET 2021**

(a)$\frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}$ (b) 4, 8, 16, 32
(c)$4\sqrt{2}, 8, 16\sqrt{2}, 32$ (d) None of these

37. If $a_1, a_2 \dots \dots a_n$ be arithmetic progression with common difference d, then the sum of $\sin d(\operatorname{cosec} a_1 \operatorname{cosec} a_2 + \operatorname{cosec} a_2 \operatorname{cosec} a_3 + \cdots + \operatorname{cosec} a_{n-1} \operatorname{cosec} a_n)$ is equal to **NIMCET 2022**
(a)$\cot a_1 - \cot a_n$ (b)$\sin a_1 - \sin a_n$
(c)$\operatorname{cosec} a_1 - \operatorname{cosec} a_n$ (d)$a_1 - a_0$

38. If $a_1, a_2, \dots a_n$ are any real number and n is any positive integer, then **NIMCET 2022**
(a) $n\sum_{i=1}^{n} a_i^2 < (\sum_{i=1}^{n} a_i)^2$ (b) $n\sum_{i=1}^{n} a_i^2 \geq (\sum_{i=1}^{n} a_i)^2$
(c) $\sum_{i=1}^{n} a_i^2 \geq (\sum_{i=1}^{n} a_i)^2$ (d) None of these

39. In a harmonic progression, p^{th} term is q and q^{th} term is p. then, pq^{th} term is **NIMCET 2022**
(a)0 (b)1
(c)pq (d)$pq(p+q)$

40. If α and β are the roots of $x^2 - x - 1 = 0$, and $A_n = \alpha^n + \beta^n$, then arithmetic mean of A_{n-1} and A_n is **NIMCET 2022**
(a)$2A_n - 1$ (b)$\frac{1}{2}A_{n+1}$
(c)$2A_n - 2$ (d)None

41. Which term of the series $\frac{\sqrt{5}}{3}, \frac{\sqrt{5}}{4}, \frac{1}{\sqrt{5}}, \dots$ is $\frac{\sqrt{5}}{13}$? **NIMCET 2022**
(a)12 (b)11
(c)10 (d)9

42. The sum of infinite terms of decreasing GP is equal to the greatest value of the function $f(x) = x^3 + 3x - 9$ in the interval [-2, 3] and difference between the first two terms is $f'(0)$. Then, the common ratio of the GP is **NIMCET 2023**
(a)$-\frac{2}{3}$ (b)$\frac{4}{3}$ (c)$+\frac{2}{3}$ (d)$-\frac{4}{3}$

43. If a, b, c, d are in HP and arithmetic mean of ab, ac, cd is 9 then which of the following number is the value of ad? **NIMCET 2023**
(a)3 (b)9 (c)12 (d)4

44. The coefficient of x^{50} in the expression of $(1+x)^{1000} + 2x(1+x)^{999} + 3x^2(1+x)^{998} + \cdots + 1001x^{1000}$ **NIMCET 2023**
(a) ${}^{1005}C_{50}$ (b) ${}^{1005}C_{48}$
(c) ${}^{1002}C_{50}$ (d) ${}^{1002}C_{51}$

45. If $x_k = \cos\left(\frac{2\pi k}{n}\right) + i\sin\left(\frac{2\pi k}{n}\right)$ then $\sum_{k=1}^{n} x_k$ =? **NIMCET 2023**
(a)1 (b)-1
(c)0 (d)None

46. The number of solutions of $5^{1+|\sin x|+|\sin x|^2+\cdots} = 25$ For x ∈ (-π, π) is (Sequence and series)
(a) 2 (b) 0
(c) 4 (d) infinite
NIMCET 2024

47. If one AM (Arithmetic mean) 'a' and two GM's (Geometric means) p and q be inserted between any two positive numbers, the value of $p^3 + q^3$ is (Sequence Series)
(a) 2apq (b) pq/a
(c) 2pq/a (d) p + q +a
NIMCET 2024

48. The value of series $\frac{2}{3!} + \frac{4}{5!} + \frac{6}{7!} + \cdots$ ……… is (Sequence Series)
(a) $2e^{-2}$ (b) e^{-2}
(c) e^{-1} (d) $2e^{-1}$
NIMCET 2024

49. The value of $\sum_{r=1}^{n} \frac{1}{2^n} \frac{{}^nP_r}{r!}$ is: (Sequence Series)
(a) 2^n (b) $1-2^{-n}$
(c) 2^n-1 (d) $2^{2n}-1$
NIMCET 2024

50. Suppose $t_1, t_2, t_3, \dots, t_{55}$ are in AP such that $\sum_{i=0}^{18} t_{3i+1} = 1197$ and $t_7 + 3t_{22} = 174$. If $\sum_{i=1}^{9} t_i^2 = 947b$, then the value of b is **NIMCET 2025**
(a) 5 (b) 3
(c) 1 (d) 2

51. A group of 630 children is arranged in rows for a group photograph. Each row contains three fewer children than the in front of it. What number of rows is not possible? **NIMCET 2025**
(a) 6 (b) 3
(c) 5 (d) 4

52. The number of all even integers between 99 and 999 which are not multiple of 3 and 5 is **NIMCET 2025**
(a) 240 (b) 250
(c) 235 (d) 245

ANSWER KEY

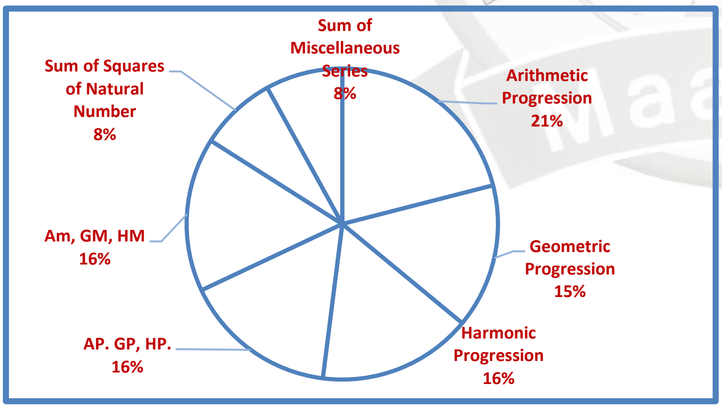
1.	a	2.	a	3.	c	4.	b	5.	c
6.	c	7.	b	8.	b	9.	c	10.	c
11.	a	12.	b	13.	d	14.	d	15.	a
16.	c	17.	d	18.	a	19.	c	20.	b
21.	a	22.	b	23.	b	24.	a	25.	a
26.	a	27.	b	28.	c	29.	c	30.	b
31.	a	32.	c	33.	d	34.	a	35.	a
36.	b	37.	a	38.	b	39.	b	40.	b
41.	b	42.	c	43.	b	44.	c	45.	c
46.	c	47.	a	48.	c	49.	b	50.	b
51.	a	52.	a						

Analysis 2007 to 2025

Questions Asked in NIMCET 2007 to 2025

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	03	NIMCET 2016	02
NIMCET 2008	03	NIMCET 2017	02
NIMCET 2009	00	NIMCET 2018	04
NIMCET 2010	01	NIMCET 2019	05
NIMCET 2011	03	NIMCET 2020	01
NIMCET 2012	02	NIMCET 2021	02
NIMCET 2013	06	NIMCET 2022	05
NIMCET 2014	00	NIMCET 2023	04
NIMCET 2015	02	NIMCET 2024	04
		NIMCET 2025	03

SUB TOPIC WISE ANALYSIS



Sub Topicwise Questions Asked in NIMCET

Sub Topic	Ques. Asked	Sub Topic	Ques. Asked
Arithmetic Progression	08	Sub Topic	
Geometric Progression	08	Sum of Miscellaneous Series	03
Harmonic Progression	07		
AP, GP, HP	06		
AM, GM, HM	06		
Sum of Squares of Natural Number	03		

SOLUTION

1. **(a)** The angle of a triangle are in AP. If b : c = $\frac{\sqrt{3}}{\sqrt{2}}$ then angle A is equal to

	A	B	C
Let Angle are –	60-d,	60,	60+d

$\frac{b}{c}=\frac{\sqrt{3}}{\sqrt{2}}$ b = K sin B

C = K sin C

$\frac{K\sin B}{K\sin C}=\frac{\sqrt{3}}{\sqrt{2}}$

$\frac{\sin 60}{\sin(60+d)}=\frac{\sqrt{3}}{\sqrt{2}}$

$\frac{\sqrt{3}/2}{\sin(60+d)}=\frac{\sqrt{3}}{\sqrt{2}}$

Sin (60 +d) $=\frac{1}{\sqrt{2}}$

Sin (60 +d) = sin 45^0

60 + d = 45

D = –15

A = 60 – (–15)

A = 75^0

2. **(a)** mth term of HP = n

Mth term of AP = 1/n

Nth term of HP = m

Nth term of AP = 1/m

Let a is first term d is common diff.

T_n = a + (n –1) d = 1/m ....(1)

T_m = a + (m–1) d = 1/n ....(2)

T_n–T_m

$\frac{1}{m}-\frac{1}{n}$ = d(n-x-m+x)

$\frac{n-m}{mn}=d(n-m)$

d = $\frac{1}{mn}$

a = $\frac{1}{m}-(n-1)\frac{1}{mn}=\frac{n-n+1}{mn}=\frac{1}{mn}$

T_{m+n} = a + (m + n –1)

$=\frac{1}{m}+(m+n-1)\times\frac{1}{mn}$

$T_{m+n}=\frac{m+n}{mn}\to AP$

$T_{m+n}=\frac{mn}{m+n}$ ....NP Ans.

3. **(c)** Let age of all student = a, a+d, a +2d

a = 5

a + 2d = 15

5 + 2d = 15

d = 5

Ages = 5, 10, 15

Average = $\frac{5+10+15}{3}$ = 10 year

4. **(b)** Given, $e^{1/c},e^{b/ac},e^{1/a}$

a,b,c are in AP $\Rightarrow 2b=c+a$

In case of exponential, first we assume that it will be in G.P.

$\Rightarrow\ e^{1/c}\times e^{1/a}=\left(e^{b/ac}\right)^2$

$\Rightarrow\ e^{\frac{1}{c}+\frac{1}{a}}=e^{\frac{2b}{ac}}$

$\Rightarrow\ e^{\frac{a+c}{ac}}=e^{\frac{2b}{ac}}$ as, given $2b=a+c$

$\Rightarrow\ e^{\frac{2b}{ac}}=e^{\frac{2b}{ac}}$

$\therefore\ e^{1/c},e^{b/ac},e^{1/a}$ in GP.

5. **(c)** , $H_1, H_2 \dots\dots\dots H_n, b$ in HP

$\Rightarrow \frac{1}{a}, \frac{1}{H_1}, \frac{1}{H_2} \cdots \frac{1}{H_n}, \frac{1}{b}$ in AP.

$\Rightarrow \frac{1}{b} = \frac{1}{a} + (n+2-1)d$

$\Rightarrow \frac{1}{b} - \frac{1}{a} = (n+1)d \Rightarrow d = \frac{\frac{1}{b}-\frac{1}{a}}{(n+1)}$

$\therefore \frac{1}{H_1} = \frac{1}{a} + d = \frac{1}{a} + \frac{\frac{1}{b}-\frac{1}{a}}{n+1} = \frac{1}{a} + \frac{a-b}{ab(n+1)}$

$\Rightarrow \frac{H_1}{a} = \frac{bn+b}{a+bn}$

$\Rightarrow$ Using componendo and dividendo, we get

$\frac{H_1+a}{H_1-a} = \frac{bn+b+bn+a}{bn+b-bn-a} = \frac{a+b+2bn}{b-a}$

Also, $\frac{1}{H_n} = \frac{1}{b} - d = \frac{1}{b} - \frac{a-b}{ab(n+1)} = \frac{a(n+1)-a+b}{ab(n+1)}$

$\Rightarrow H_n = \frac{ab(n+1)}{an+b} \Rightarrow \frac{H_n}{b} = \frac{an+a}{an+b}$

Using componendo and dividendo, we get

$\Rightarrow \frac{H_n+b}{H_n-b} = \frac{an+a+an+b}{an+a-an-b} = \frac{a+b+2an}{a-b}$

On adding Eqs. (i) and (ii),

$\frac{H_1+a}{H_1-a} + \frac{H_n+b}{H_n-b} = \frac{a+b+2bn}{b-a} - \frac{a+b+2an}{b-a}$

$\Rightarrow \frac{2n(b-a)}{b-a} = 2n$

6. **(c)** $1+2+3+\cdots+n = \frac{n\times(n+1)}{2}$

$\bar{x} = \frac{1+2+3+\cdots+n}{n}$

$\bar{x} = \frac{n\times(n+1)}{2\times n}$

$\bar{x} = \frac{(n+1)}{2}$

$\bar{x} = \frac{(n+7)}{3}$ (given)

Hence, $\frac{n+1}{2} = \frac{n+7}{3}$

$\Rightarrow 3(n+1) = 2(n+7) \Rightarrow 3n+3 = 2n+14$

n $= 14-3 = 11$

7. **(b)** (b) a, b, c are in AP,

$2b = a+c \Rightarrow (2b)^2 = (a+c)^2 \Rightarrow 4b^2 = (a+c)^2$

p, q, r are in HP,

$q = \frac{2pr}{p+r} \Rightarrow p+r = \frac{2pr}{q}$

and ap, bq, cr are in GP.

$(bq)^2 = (ap)(cr) \Rightarrow b^2q^2 = (ap)(cr)$

$\Rightarrow b^2\left(\frac{2pr}{p+r}\right)^2 = acpr \Rightarrow \frac{4b^2p^2r^2}{(p+r)^2} = acpr$

$\Rightarrow \frac{4b^2pr}{(p+r)^2} = ac \Rightarrow \frac{(p+r)^2}{4pr} = \frac{b^2}{ac} = \frac{(a+c)^2}{4ac}$

$\Rightarrow \frac{p^2+r^2+2pr}{pr} = \frac{a^2+c^2+2ac}{ac}$

$\Rightarrow \frac{p}{r} + \frac{r}{p} + 2 = \frac{a}{c} + \frac{c}{a} + 2 \Rightarrow \frac{p}{r} + \frac{r}{p} = \frac{a}{c} + \frac{c}{a}$

8. (b) $11^2 + 12^2 + \cdots + 30^2$

$(1^2+2^2+\cdots 30^2) - (1^2+2^2+\cdots+10^2)$

$= \frac{30(30+1)(60+1)}{6} - \left[\frac{10\times(10+1)(20+1)}{6}\right]$

$\left[\Sigma n^2 = \frac{n(n+1)(2n+1)}{6}\right]$

$= \frac{(930)(60+1)}{6} - \left[\frac{(110)(20+1)}{6}\right]$

$= \frac{56730}{6} - \frac{2310}{6} = \frac{54420}{6} = 9070$

9. **(c)** $\log(a+c) + \log(a-2b+c)$

Since, a, b, c in HP.

$\therefore b = \frac{2ac}{a+c}$

$= \log(a+c) + \log(a-2b+c)$

$= \log(a+c) + \log((a+c)-2b)$

$= \log[(a+c)^2 - 2b(a+c)]$

$= \log\left[(a+c)^2 - 2\frac{2ac}{a+c}(a+c)\right]$

$= \log(a-c)^2 = 2\log|(a-c)| = 2\log|(c-a)|$

10. **(c)** $1^2+2^2+3^2+\cdots+n^2$ be the series (say)

Using average mean formula, $\bar{x} = \frac{x_1+x_2+\cdots+x_n}{n}$

$\bar{x} = \frac{1}{n}\{1^2+2^2+3^2+\cdots+n^2\}$ (according to the question)

$11 = \frac{1}{n}\left\{\frac{n(n+1)(2n+1)}{6}\right\}\left[\because \Sigma n^2 = \frac{n(n+1)(2n+1)}{6}\right]$

$11 = \frac{2n^2+n+2n+1}{6} \Rightarrow 2n^2+3n+1 = 66$

$\Rightarrow 2n^2+13n-10n-65 = 0$

$\Rightarrow n(2n+13) - 5(2n+13) = 0$

$\Rightarrow (n-5)(2n+13) = 0$

$\Rightarrow n-5 = 0$ or $2n+13 = 0$

$\Rightarrow n = 5$ or $n = \frac{-13}{2}$

Since, n can't be negative, hence n = 5.

11. **(a)** H is the harmonic mean between P and Q (given)

$H = \frac{2PQ}{P+Q} \Rightarrow \frac{H}{2} = \frac{PQ}{P+Q}$

$\frac{H}{P} + \frac{H}{Q} = H\left(\frac{P+Q}{PQ}\right) = H\left(\frac{2}{H}\right) = 2$

12. **(b)** $F(x) = x + \frac{1}{x}$

For $x > 0$

Using $AM \geq GM$

$x + \frac{1}{x} \geq 2$

Hence, minimum value of $x + \frac{1}{x} = 2$

At $x = 1$, the value of $x + \frac{1}{x} = 2$

13. (d) Let the term of GP be $a, ar, ar^2, \ldots..$

According to question,

$a = ar + ar^2 \Rightarrow a = a(r + r^2)$

$r^2 + r - 1 = 0$

$r = \frac{-1\pm\sqrt{1+4}}{2} \Rightarrow r = \frac{-1\pm\sqrt{5}}{2}$

Since, ' r ' is positive.

Hence,

$r = \frac{-1+\sqrt{5}}{2}$

14. (d) $S_n = 216$ (given)

$S_n = \frac{n}{2}[2a + (n-1)d]$

$\Rightarrow 216 = \frac{n}{2}[2a + (n-1)d]$

$T_n = 2n, a = n$

$2n = n + (n-1)d$

$\Rightarrow d = \frac{n}{(n-1)}$

Then, $216 = \frac{n}{2}\left[2n + (n-1) \times \frac{n}{n-1}\right]$

$\frac{n}{2}[3n] = 216 \Rightarrow \frac{3n^2}{2} = 216$

$\Rightarrow n^2 = \frac{216\times2}{3} \Rightarrow n^2 = 72 \times 2 \Rightarrow n^2 = 144 \Rightarrow n = 12$

Putting in Eq. (i)

$d = \frac{12}{11}$

15. (a) First term multiple of 7 in between 200 and 400 is 203

$203 + 210 + \cdots + 399$

Let 399 is nth term of AP, then

$T_n = a + (n-1) \times d \Rightarrow 399$

$= 203 + (n-1) \times 7$

$n \Rightarrow \frac{399-203}{7} + 1 = 29$ terms in the series

Sum of 29 terms $= \frac{n}{2}(a + l) = 29 \times \frac{(203+399)}{2} = 8729$

16. (c) $-1^2 + 2^2 - 3^2 + 4^2 - \cdots (20)^2$ (given)

$\Rightarrow \{(2)^2 - (1)^2\} + \{(4)^2 - (3)^2\} + \cdots + \{(20)^2 - (19)^2\}$

$\Rightarrow 3 + 7 + 11 + \cdots + 39$

Required sum for (10 terms)

$\frac{n}{2}(a + l) = 10 \times \left(\frac{3+39}{2}\right) = 210$

17. (d) $f(x) = lx^2 + mx + n$

As

$f(1) = f(-1)$

$l + m + n = l - m + n$

Hence, $m = 0$

Now, $f(x) = lx^2 + n$

$f'(x) = 2lx \Rightarrow f'(a) = 2la$

$f'(b) = 2lb, \quad f'(c) = 2lc$

As a, b, c are in AP, hence $f'(a), f'(b), f'(c)$ are in AP.

18. (a) $9^{\frac{1}{3}} \cdot 9^{\frac{1}{9}} \cdot 9^{\frac{1}{27}} \ldots = 9^{\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}\cdots} = 9^{\frac{1/3}{1-\frac{1}{3}}} = 9^{\frac{1}{2}} = 3$

19. (c) If a, b, c are in geometric progression, then $b^2 = ac$ so by multiplying both sides by x^2,

$x^2b^2 = x^2ac \Rightarrow x^2b^2 = (xa)(xc)$

Taking log both sides of base x

$\log_x(x^2b^2) = \log_x[(xa)(xc)]$

$\Rightarrow 2\log_x(xb) = \log_x xa + \log_x xc$

Hence, $\log_x ax, \log_x bx, \log_x cx$ are in AP.

$\frac{1}{\log_{ax} x}, \frac{1}{\log_{bx} x}, \frac{1}{\log_{cx} x}$ are in AP

Hence, $\log_{ax} x, \log_{bx} x, \log_{cx} x$ are in HP.

20. (b) $\frac{1}{2\sqrt{1}+1\sqrt{2}} + \frac{1}{3\sqrt{2}+2\sqrt{3}} + \frac{1}{4\sqrt{3}+3\sqrt{4}} + \cdots \ldots + \frac{1}{25\sqrt{24}+24\sqrt{25}}$

Its general term $T_n = \frac{1}{(n+1)\sqrt{n}+n\sqrt{n+1}}$

$= \frac{1}{(n+1)\sqrt{n}+n\sqrt{n+1}} \times \frac{(n+1)\sqrt{n}-n\sqrt{n+1}}{(n+1)\sqrt{n}-n\sqrt{n+1}}$

$= \frac{(n+1)\sqrt{n}-n\sqrt{n+1}}{n(n+1)^2-n^2(n+1)} = \frac{(n+1)\sqrt{n}-n\sqrt{n+1}}{n(n+1)}$

$T_n = \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}$

Sum, $S = \sum_{n=1}^{24} T_n = \sum_{n=1}^{24}\left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}\right)$

$S = \left(\frac{1}{1} - \frac{1}{\sqrt{2}}\right) + \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{3}}\right) + \left(\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}}\right) + \cdots$

$+\left(\frac{1}{\sqrt{24}} - \frac{1}{\sqrt{25}}\right)$

$S = \frac{1}{1} - \frac{1}{\sqrt{25}} \Rightarrow S = 1 - \frac{1}{5} \Rightarrow S = \frac{4}{5}$

21. (a) $a_1, a_2, a_3, \ldots \ldots, a_n$ are in AP (given)

$a_1 = 0$

So, $a_2 = a_1 + d = d$

$a_3 = a_1 + 2d = 2d$

$a_{n-1} = (n-2)d \Rightarrow a_n = (n-1)d$

$\therefore \left(\frac{a_3}{a_2} + \frac{a_4}{a_3} + \cdots . + \frac{a_n}{a_{n-1}}\right) - a_2\left(\frac{1}{a_2} + \frac{1}{a_3} + \cdots + \frac{1}{a_{n-2}}\right)$

$(n-2)$ terms $(n-3)$ terms

$= \left[\left(\frac{2d}{d}\right) + \frac{3d}{2d} + \cdots + \frac{(n-1)d}{(n-2)d}\right] - d_2\left(\frac{1}{d} + \frac{1}{2d} + \cdots + \frac{1}{(n-3)d}\right)$

$= \left[2 + \frac{3}{2} + \cdots . + \frac{(n-1)}{(n-2)}\right] - \left[1 + \frac{1}{2} + \cdots . + \frac{1}{n-3}\right]$

$= 1 + 1 + 1 \ldots (n-2)$ times $+\left\{1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{(n-3)} + \frac{1}{(n-2)}\right\}$

$-\left\{1+\frac{1}{2}+\frac{1}{3}+\cdots+\frac{1}{(n-3)}\right\}$

$=(n-2)+\dfrac{1}{(n-2)}$

22. (b) Let

$S=\frac{1}{\sqrt{1}+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\frac{1}{\sqrt{3}+\sqrt{4}}+\cdots+\frac{1}{\sqrt{80}+\sqrt{81}}$

After rationalisation rule in each term, we get

S = $\frac{\sqrt{2}-\sqrt{1}}{2-1}+\frac{\sqrt{3}-\sqrt{2}}{3-2}+\cdots+\frac{\sqrt{81}-\sqrt{80}}{81-80}$

$=\sqrt{2}-1+\sqrt{3}-\sqrt{2}+\cdots+\sqrt{81}-\sqrt{80}=\sqrt{81}-1$

$9-1=8$

23. (b) Let the two numbers are x and y.

Given, the arithmetic mean and geometric mean of the x and y is A and G

$\Rightarrow A=\frac{x+y}{2}$ .... (i)

$\Rightarrow G^2=xy$ .... (ii)

The harmonic mean at two number x and y is 4 (given)

$\Rightarrow \frac{2xy}{x+y}=4$

$\Rightarrow 2xy=4(x+y)$

$\Rightarrow xy=2(x+y)$

$\Rightarrow G^2=4A$ [using Eqs. (i) and (ii)]

Given, their arithmetic mean A and the geometric mean G satisfy the relation $2A+G^2=27$.

$\Rightarrow 2A+G^2=27\ [\because G^2=4A]$

$\Rightarrow 6A=27$

$\Rightarrow A=\frac{27}{6}$

$\Rightarrow A=\frac{9}{2}$

From Eqs. (i) and (ii), we have

$x+y=9$ and $xy=18$

$\Rightarrow x=6$ and $y=3$

24. (a) Consider three positive numbers $a-d, a, a+d$ are in AP.

Three positive numbers whose sum is 21

$\Rightarrow a-d+a+(a+d)=21$

⇒ a = 7

Hence, three positive numbers become $7-d, 7, 7+d$

It is given, if 2,2,14 are added to them resulting number are in GP.

$9-d, 9, 21+d$ are in GP.

$\Rightarrow 81=(21+d)(9-d)$

$\Rightarrow d^2+12d-108=0$

$\Rightarrow (d+18)(d-6)=0$

$\Rightarrow d=-18$ and $d=6$

When $d=-18$, the three numbers are $25, 7, -11$. (given that numbers which are in AP, all are positive, hence these numbers $25, 7, -11$ are rejected)

When $d=6$, the three numbers are 1,7,13

25. (a) Given that $a_1, a_2, \ldots, a_n$ be positive real number such that $a_1, a_2 \ldots a_{n-1}a_n=C$

We know that AM $\geq$ GM for positive numbers.

We apply AM,GM concept on $a_1, a_2, a_3, a_4 \ldots a_{n-1}, 2a_n$

$\frac{a_1+a_2+\cdots+a_{n-1}+2a_n}{n}\geq\left(a_1a_2\ldots a_{n-1}(2a_n)\right)^{\frac{1}{n}}$

$\Rightarrow a_1+a_2+\cdots+a_{n-1}+2a_n\geq n(2C)^{1/n}$

So, minimum value are

$a_1+a_2+\cdots+a_{n-1}+2a_n$ is $n(2C)^{1/n}$

26. (a) Let the first term of the GP be a and common ratio r.

$\Rightarrow \frac{a}{1-r}=2(a+ar)=2a(1+r)\Rightarrow(1+r)(1-r)=\frac{1}{2}$

$\Rightarrow 1-r^2=\frac{1}{2}\Rightarrow r^2=\frac{1}{2}$

$\Rightarrow r=\pm\frac{1}{\sqrt{2}}$

Hence, the possible values of the common ratio are $\frac{1}{\sqrt{2}}$ and $\frac{-1}{\sqrt{2}}$

27. (b) In question it should be given that these terms are in H.P.

Given that m^{th} term of HP

$\frac{1}{n}=\frac{1}{a}+(m-1)d$

And

n^{th} term $\frac{1}{m}=\frac{1}{a}+(n-1)d$

Substract Eq. (ii) from Eq. (i), we get

$\frac{1}{n}-\frac{1}{m}=d(m-n)$

Hence, $d=\frac{1}{mn}$

Put this value in Eq. (i), we get $a=mn$

$T_{m+n}=\frac{1}{\frac{1}{mn}+(m+n-1)\frac{1}{mn}}=\frac{mn}{m+n}$

28. (c) Given that mean of the squares of first n natural numbers be 11 .

$\Rightarrow \frac{n(n+1)(2n+1)}{6n}=11\Rightarrow(n+1)(2n+1)=66$

$\Rightarrow 2n^2+3n+1=66\Rightarrow 2n^2+3n-65=0$

$\Rightarrow n=\frac{-3\pm\sqrt{9+520}}{4}\Rightarrow n=\frac{-3\pm\sqrt{529}}{4}$

$\Rightarrow n=\frac{-3+23}{4}=\frac{20}{4}=5$ [Taking the $+$ve sign]

As, $n=\frac{-3-23}{4}=\frac{-26}{4}$ [n cannot be negative]

$\therefore\ n=5$

29. **(c)** Clearly a_n, b_n, c_n are the middle terms of given AP, GP and HP, respectively.

So, $a_n=$ AM of a and b

$b_n=$ GM of a and b

$c_n=$ HM of a and b

Since, AM, GM and HM are in GP $\Rightarrow G^2=AH$

$\therefore\ b_n^2=a_nc_n$

Now, given equation is $a_nx^2-b_nx+c_n=0$

$\therefore\ D=b_n^2-4a_nc_n$

$\Rightarrow\ D=a_nc_n-4a_nc_n \Rightarrow D=-3a_nc_n$

$D<0(\because a_n, c_n>0$ as a and b are positive)

So, given equation has imaginary roots.

30. **(b)** a, b, c are in GP $\Rightarrow b^2=ac$

Also, $\log a-\log 2b, \log 2b-\log 3c, \log 3c-\log a$ are in AP.

$\Rightarrow 2(\log 2b-\log 3c)=\log a-\log 2b+\log 3c-\log a$

$\Rightarrow 3\log 2b=3\log 3c \Rightarrow \log 2b=\log 3c$

$\Rightarrow 2b=3c \Rightarrow b=\frac{3c}{2}$

Now, $b^2=ac$

$\Rightarrow \frac{9}{4}c^2=ac \Rightarrow a=\frac{9}{4}c$

$\because\ a$ is greatest side

$\cos A=\frac{b^2+c^2-a^2}{2bc}=\frac{\frac{9}{4}c^2+c^2-\frac{81}{16}c^2}{2\times\frac{3}{2}c^2}<0$

$\therefore \angle A$ is obtuse angle.

So, $\triangle ABC$ is obtuse anlged triangle.

31. **(a)** $x, 2x+2, 3x+3$ are in GP

$\therefore\ (2x+2)^2=x(3x+3)$

$\Rightarrow\ 4(x+1)^2=3x(x+1) \Rightarrow x=-1,-4$

Put $x=-1$, series will be $-1,0.0$ (we should know that no any term of GP is zero)

Put $x=-4$ in GP

$\therefore$ The GP is $-4,-6,-9, \ldots.$

$a_4=-4\left(\frac{3}{2}\right)^3=-4\times\frac{27}{8}=\frac{-27}{2}=-13.5$

32. **(c)** $f(x)=x^3+3x-9, x\in[-2,3]$

Differentiate with respect to x.

$f'(x)=3x^2+3$

Hence, $f(x)$ is strictly increasing function, so its greatest value will be at $x=3$

$f(3)=3^3+3\times3-9=27$

$\frac{a}{1-r}=27 \Rightarrow a=27-27r$

$a+27r=27$

$f'(0)=3$

Also, given $a-ar=f'(0) \Rightarrow a(1-r)=3$

$\Rightarrow\ 1-r=\frac{3}{a}$

From Eqs. (i) and (ii), we get

$a+27\left(1-\frac{3}{a}\right)=27$

$\Rightarrow\ a+27-\frac{81}{a}=27$

$\Rightarrow\ a^2=81$

$\Rightarrow\ a=\pm9$

$\therefore$ GP is decreasing

$\therefore a=9$

Now, $\frac{9}{1-r}=27 \Rightarrow \frac{1}{3}=1-r$

$\Rightarrow\ r=1-\frac{1}{3} \Rightarrow r=\frac{2}{3}$

33. **(d)** $\Sigma x_i=80$ and $\Sigma x_i^2=400$

Let $x_1, x_2 \ldots x_n$ be n observations.

We know,

Arithmetic mean of m^{th} power $\geq m^{\text{th}}$ power of arithmetic mean

$\frac{x_1^m+x_2^m+\cdots+x_n^m}{n}\geq\left(\frac{x_1+x_2+\cdots+x_n}{n}\right)^m$

Taking $m=2$

$\frac{x_1^2+x_2^2+\cdots+x_n^2}{n}\geq\left(\frac{x_1+x_2+\cdots x_n}{n}\right)^2$

$\Rightarrow \frac{\Sigma x_i^2}{n}\geq\left(\frac{\Sigma x_i}{n}\right)^2$

$\Rightarrow \frac{400}{n}\geq\left(\frac{80}{n}\right)^2 \Rightarrow \frac{400}{n}\geq\frac{6400}{n^2} \Rightarrow n\geq16$

As n is a positive integer from given option (d) is the right answer.

34. **(a)** It is given that first term $a=3$,

let common difference $=d$ (say)

Also the sum of first 8 terms is twice the sum at the first 5 terms.

$\Rightarrow S_8=2S_5$

$\Rightarrow \frac{8}{2}[6+(8-1)d]=2\times\frac{5}{2}[6+(5-1)d]$

$\Rightarrow 4(6+7d)=5(6+4d)$

$\Rightarrow 24+28d=30+20d$

$\Rightarrow$ 8d = 6

$\Rightarrow$ d $=\frac{3}{4}$

$\therefore$ Common difference that AP is 3/4.

35. (a) Refer to solution 2.

36. (b) Let four geometric mean between 2 and 64 are G_1, G_2, G_3, G_4.

$\therefore$ $2, G_1, G_2, G_3, G_4, 64$ form a GP.

First term $(a) = 2$ and sixth term $= 64$

$ar^5 = 64, r =$ common ratio

$2 \times r^5 = 64 \Rightarrow r^5 = 32 \Rightarrow r = 2$

$G_1 = ar = 2 \times 2 = 4$

$G_2 = ar^2 = 2(2)^2 = 8$

$G_3 = ar^3 = 2(2)^3 = 16$

$G_4 = ar^4 = 2(2)^4 = 32$

37. (a) We have

$d = a_2 - a_1 = a_3 - a_2 = a_4 - a_3 = \cdots = a_n - a_{n-1}$

Thus, $\sin d[\operatorname{cosec} a_1 \operatorname{cosec} a_2 + \operatorname{cosec} a_2 \operatorname{cosec} a_3 + \cdots + \operatorname{cosec} a_{n-1} \operatorname{cosec} a_n]$

$= \frac{\sin(a_2-a_1)}{\sin a_1 \sin a_2} + \frac{\sin(a_3-a_2)}{\sin a_2 \sin a_3} + \cdots + \frac{\sin(a_n-a_{n-1})}{\sin a_{n-1} \sin a_n}$

$= (\cot a_1 - \cot a_2) + (\cot a_2 - \cot a_3) + \cdots + \cot a_1 - \cot a_n\ (\cot a_{n-1} - \cot a_n)$

Hence, option (a) is the correct answer.

38. (b) Let $a_1, a_2, a_3, \ldots. a_n$ are n positive numbers

AM of m^{th} power of numbers $\geq (\text{AM})^m$

$\frac{a_1^m + a_2^m + a_3^m + \cdots + a_n^m}{n} \geq \left(\frac{a_1 + a_2 + a_3 + \cdots + a_n}{n}\right)^m$

m =2

$\frac{a_1^2 + a_2^2 + a_3^2 + \cdots. + a_n^2}{n} \geq \left(\frac{a_1 + a_2 + a_3 + \cdots + a_n}{n}\right)^2$

$\Rightarrow n\sum_{i=1}^n a_i^2 \geq (\sum_{i=1}^n a_i)^2$

39. (b) H.P. A.P.

$T_p = q\ T_P = 1/q = A + (p-1)d$

$T_q = p\ T_q = 1/p = A + (q-1)d$

$\frac{1}{q} - \frac{1}{p} = (p-q)d \Rightarrow d = \frac{1}{pq}$

Put this value in Eq. (ii)

$\frac{1}{p} = A + (q-1)\frac{1}{pq}$

$= A + \frac{1}{p} - \frac{1}{pq}$

$\Rightarrow A = \frac{1}{pq}$

$T_{pq} = A + (pq-1)d = \frac{1}{pq} + (pq-1) \times \frac{1}{pq} = 1$

40. (b) If α and β are the roots of

$x^2 - x - 1 = 0$

$\alpha + \beta = 1, \alpha\beta = -1$

$A_n = \alpha^n + \beta^n$

AM of A_{n-1} and $A_n = \frac{A_{n-1} + A_n}{2}$

$= \frac{\alpha^{n-1} + \beta^{n-1} + \alpha^n + \beta^n}{2}$

$= \frac{\alpha^{n-1}(1+\alpha) + \beta^{n-1}(1+\beta)}{2}$

As α, β are roots of $x^2 - x - 1 = 0$

Hence, $\alpha^2 - \alpha - 1 = 0 \Rightarrow 1 + \alpha = \alpha^2$ and $1 + \beta = \beta^2$

$\frac{\alpha^{n-1}(\alpha^2) + \beta^{n-1}\beta^2}{2} = \frac{\alpha^{n+1} + \beta^{n+1}}{2} = \frac{A_{n+1}}{2}$

41. (b) $\frac{\sqrt{5}}{3}, \frac{\sqrt{5}}{4}, \frac{\sqrt{5}}{5}, \frac{\sqrt{5}}{6}, \frac{\sqrt{5}}{7}, \ldots \frac{\sqrt{5}}{13}$

Observe its denominators

3,4,5,6, ... to 13 → Total 11 terms

42. (c) Refer to solution 29.

43. (b) $\frac{ab+bc+cd}{3} = 9$

$\Rightarrow\ ab + bc + cd = 27$

a, b, c are in HP $\Rightarrow b = \frac{2ac}{a+c}$

$\Rightarrow\ a + c = \frac{2ac}{b}$

b, c, d are in HP $\Rightarrow c = \frac{2bd}{b+d}$

$\Rightarrow\ b + d = \frac{2bd}{c}$

Multiply Eqs. (i) and (ii)

$(a+c)(b+d) = \frac{2ac}{b}\frac{2bd}{c} = 4\text{ad}$

$\Rightarrow (ab + ad + bc + dc) = 4\text{ad}$

$\Rightarrow 3ad = ab + bc + cd = 27$

$\Rightarrow$ ad =9

44. (c) Let $S = (1+x)^{1000} + 2x(1+x)^{999} + 3x^2(1+x)^{998} + \ldots + 1000x^{999}(1+x) + 1001x^{1000}$

Above is A.G.P of common ratio $r = \frac{x}{1+x}$

$\therefore \left[\frac{x}{(1+x)}\right] S = x(1+x)^{999} + 2x^2(1+x)^{998} + \cdots \ldots + 1000 \cdot x^{1000} + \frac{1001x^{1001}}{1+x}$

Subtracting Eq. (ii) from Eq. (i)

$\left(1 - \frac{x}{1+x}\right) S = (1+x)^{1000} + x(1+x)^{999} + x^2(1+x)^{998}$

$+\cdots+x^{1000}-\frac{1001x^{1001}}{1+x}$

Or $S=(1+x)^{1001}+x(1+x)^{1000}+x^2(1+x)^{999}$

$+\cdots+x^{1000}(1+x)-1001x^{1001}$

$=\frac{(1+x)^{1001}[1-(x-(1+x))^{1001}]}{1-x}-1001x^{1001}$

Sum G.P. $(1+x)^{1002}\left[1-\left(\frac{x}{1+x}\right)^{1001}\right]-1001x^{1001}$

$=(1+x)^{1002}-x^{1001}(1+x)-1001x^{1001}$

$=(1+x)^{1002}-x^{1002}-1002x^{1001}$

Now, the coefficients of x^{50} on the R.H.S of Eq. (iii)

$= {}^{1002}C_{50}$

45. (c) We should know that

$\cos\theta+i\sin\theta=e^{i\theta}$

$x_k=\cos\left(\frac{2\pi k}{n}\right)+i\sin\left(\frac{2\pi k}{n}\right)=e^{i\frac{2k\pi}{n}}$

$\sum_{k=1}^{n}x_k=\sum_{k=1}^{n}e^{i\frac{2k\pi}{n}}=e^{i\frac{2\pi}{n}}+e^{i\frac{4\pi}{n}}+e^{i\frac{6\pi}{n}}+\cdots+e^{i\frac{2n\pi}{n}}$

Let $e^{i\frac{2\pi}{n}}=\alpha$

$$\alpha+\alpha^2+\alpha^3+\cdots+\alpha^n=\frac{\alpha(1-\alpha^n)}{1-\alpha}$$

Hence this series $=\frac{e^{\frac{i2\pi}{n}}\left(1-e^{\left(\frac{i2\pi}{n}\right)n}\right)}{1-e^{\frac{i2\pi}{n}}}$

$=\frac{e^{i\frac{2\pi}{n}}(1-e^{i2\pi})}{1-e^{\frac{i2\pi}{n}}}=0$ $[\because e^{i2\pi}=\cos 2\pi+i\sin 2\pi=1]$

46. (c) The number of solution of $5^{1+|\sin x|+|\sin x|^2+\ldots\ldots}=25$

For $x\in(-\pi,\pi)$

$5^{1+|\sin x|+|\sin x|^2+\ldots\ldots}=25\ldots\ldots=5^2$

$1+|\sin x|+|\sin x|^2\ldots\ldots=2$

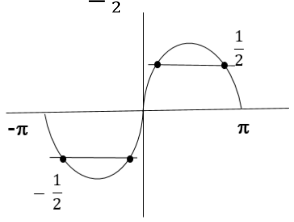
$|\sin x|+|\sin x|^2+\ldots\ldots=1$

$\frac{|\sin x|}{1-|\sin x|}=1$

$|\sin x|=1-|\sin x|$

$2|\sin x|=1$

$\sin x=\pm\frac{1}{2}$



Total 4 Solution.

47. (a) AM = a, GM's = P and q

Let 2 no. 1, 8

AM a = 9/2

GM P, q = 2,4

$P^3+q^3=8+64=72$

$P^3+q^3=2pqa$

48. (c)

Value of the series $\frac{2}{3!}+\frac{4}{5!}+\frac{6}{5!}+\cdots\ldots..$

$\sum_{n=1}^{\infty}\frac{2n+1-1}{2n!+1}$

$\sum_{n=1}^{\infty}\frac{2n+1}{2n!+1}-\frac{-1}{2n!+1}$

$\sum_{n=1}^{\infty}\frac{1}{2n!+1}-\frac{-1}{2n!+1}$

$\Rightarrow 1+1+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}+\frac{1}{5!}-\frac{1}{6!}+\cdots..$

$=e^{-1}$.

49. (b)

Yes! Here's a short and easy trick for this type of sum:

We have:

$\sum_{r=1}^{n}\frac{nP_r}{r!.2^n}$

$\frac{nP_r}{r!}=\binom{n}{r}$

So the sum becomes:

$\frac{1}{2^n}\sum_{r=1}^{n}\binom{n}{r}$

Remember a tiny binomial trick

$\sum_{r=0}^{n}\binom{n}{r}=2^n$

$\sum_{r=1}^{n}\binom{n}{r}=2^n-1$

Divide by $2n^2$

$\frac{2^n-1}{2^n}=1-\frac{1}{2^n}$

$1-2^{-n}$

50. (b)

To solve this problem, we will use the properties of an arithmetic progression (AP).

Solution:

1. Understanding the given information:

We are given that $t_1,t_2,t_3,\ldots,t_{55}$ are in an AP. Let a be the first term and d be the common difference. Then $t_n=a+(n-1)d$. We are given two equations: (i) $\sum_{l=0}^{18}t_{3l+1}=1197$ (ii) $t_7+3t_{22}=174$ We need to find b such that $\sum_{t=1}^{9}t_t^2=947b$.

1. Simplifying equation (i):

The terms in the sum are $t_1,t_4,t_7,\ldots,t_{55}$. This is an AP with first term $t_1=a$ and common difference $3d$. There are 19 terms (from $l=0$ to $l=18$). The sum of an AP is given by $S_n=\frac{n}{2}(2A+(n-1)D)$, where A is the first term and D is the common difference. So, $1197=\frac{19}{2}(2a+(19-1)3d)$ $1197=\frac{19}{2}(2a+54d)$ $1197=19(a+27d)$ $63=a+27d$ (Equation 1)

1. Simplifying equation (ii):

$t_7=a+6d$ $t_{22}=a+21d$ Substitute these into equation (ii): $(a+6d)+3(a+21d)=174$ $a+6d+3a+63d=174$ $4a+69d=174$ (Equation 2)

1. Solving the system of equations:

From Equation 1, $a=63-27d$. Substitute this into Equation 2: $4(63-27d)+69d=174$ $252-108d+69d=174$ $252-39d=174$ $39d=252-174$ $39d=78$ $d=2$ Now, find a: $a=63-27(2)=63-54=9$

1. Calculating $\sum_{t=1}^{9}t_t^2$:

The terms are $t_1,t_2,\ldots,t_9$. $t_n=9+(n-1)2$ $t_n=2n+7$ We need to calculate $\sum_{n=1}^{9}(2n+7)^2$. $\sum_{n=1}^{9}(4n^2+28n+49)=4\sum_{n=1}^{9}n^2+28\sum_{n=1}^{9}n+\sum_{n=1}^{9}49$ Using the formulas $\sum n=\frac{n(n+1)}{2}$ and $\sum n^2=\frac{n(n+1)(2n+1)}{6}$: $\sum_{n=1}^{9}n=\frac{9(10)}{2}=45$

$\sum_{n=1}^{9} n^2 = \frac{9(10)(19)}{6} = 3(5)(19) = 285$ The sum is: $4(285) + 28(45) + 9(49)$
$= 1140 + 1260 + 441 = 2841$

1. Finding the value of b:

We are given that $\sum_{t=1}^{9} t_t^2 = 947b$. So, $2841 = 947b$ $b = \frac{2841}{947} = 3$

The final answer is $\boxed{3}$.

51. (a)

In case of 3 rows, the number of children in each rows would be: n, n - 3, n - 6

⇒ n + n - 3 + n - 6 = 630

⇒ 3n - 9 = 630

⇒ 3n = 639

⇒ n = 213 (Possible)

In case of 4 rows, the number of children in each rows would be: n, n - 3, n - 6, n - 9

⇒ n = 162 (Possible)

⇒ n + n - 3 + n - 6 + n - 9 = 4n - 18 = 630

In case of 5 rows, the number of children in each rows would be:

n, n - 3, n - 6, n - 9, n - 12

⇒ n + n - 3 + n - 6 + n - 9 + n - 12 = 5n - 30 = 630

⇒ 5n = 660

⇒ n = 132 (Possible)

In case of 5 rows, the number of children in each rows would be: n, n - 3, n - 6, n - 9, n - 12, n - 15

⇒ n + n - 3 + n - 6 + n - 9 + n - 12 + n - 15 = 630

⇒ 6n - 45 = 630

⇒ 6n = 675

n = 112.5 (not possible as there cannot be 112.5 children in the first row)

Hence, we cannot form 6 rows.

52. (a)

1. Total even numbers between 99 and 999

100 to 998:

$$\frac{998-100}{2} + 1 = 450$$

2. Multiples to remove

- Even multiples of 3 = multiples of 6: from 102 to 996 →

$\frac{(996-102)}{6} + 1 = 150$

- Even multiples of 5 = multiples of 10: from 100 to 990 →

$$\frac{990-100}{10} + 1 = 90$$

- Even multiples of 3 & 5 = multiples of 30: from 120 to 990 →

$$\frac{(990-120)}{30} + 1 = 30$$

3. Inclusion-Exclusion:

$$450 - (150 + 90 - 30) = 450 - 210 = 240$$